《Automotive Configuration》

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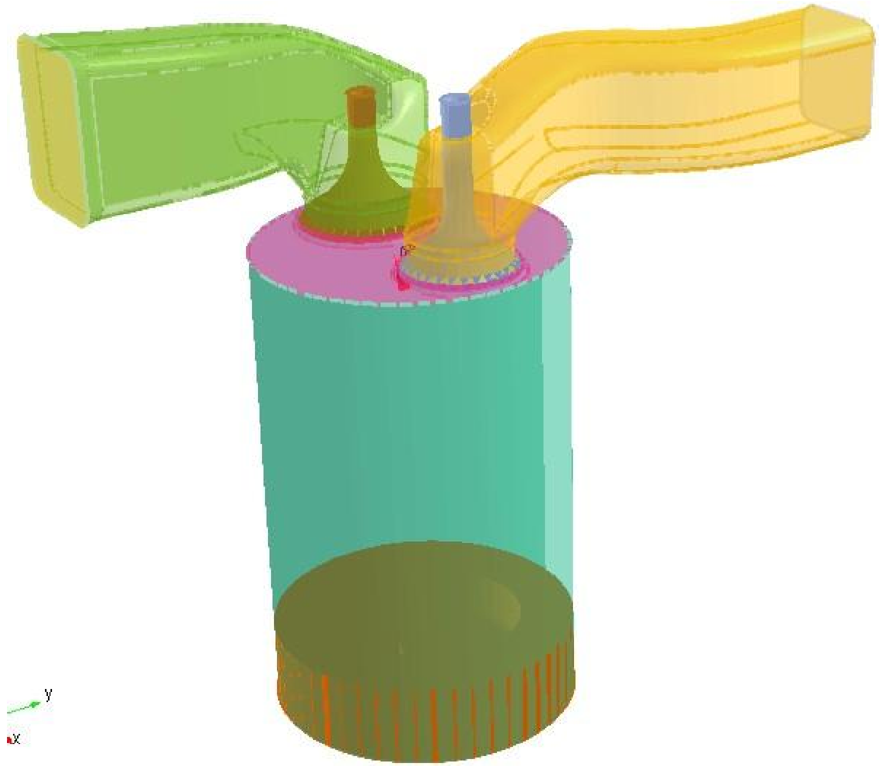
Lecture Schedule： 2nd Semester Week 1~8

Chapter 6 Intake, Exhaust and Boost Systems
进气、排气及增压系统

6.1 Intake System 进气系统

6.2 Exhaust System 排气系统

6.3 Boost System 增压系统



6.1 Intake System 进气系统

For engines equipped with electric control fuel system, intake system consists of air filter, intake port, intake manifold and mass air flow meter or intake port pressure sensor.

电喷发动机中，进气系统包括空气滤清器、进气总管、进气歧管、空气流量计或进气管压力传感器等。

[Carburetor intake system overview 化油器进气系统概述（Video）]

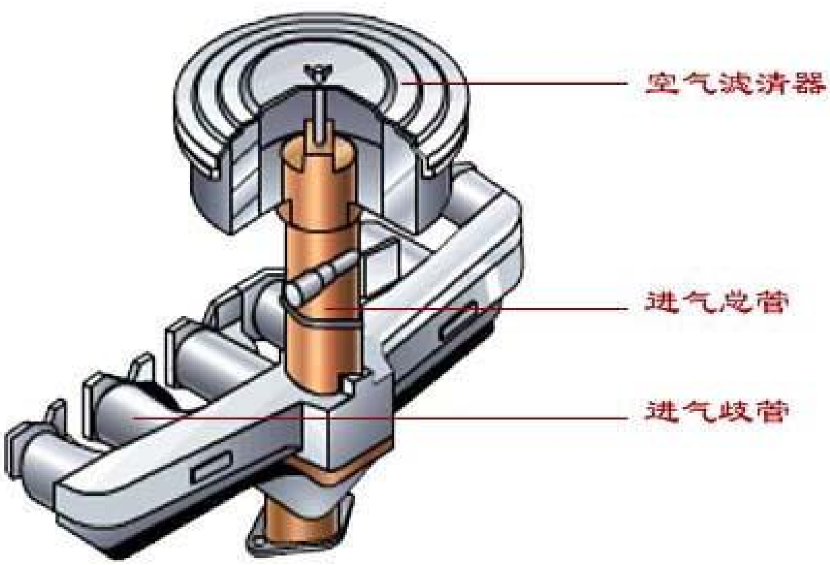
[Diesel engine intake system overview 柴油机进气系统概述（Video）]

1. Air filter 空气滤清器

Air filter comprises air intake guide tube， air filter cover， air filter casing, filter cartridge, etc.

一般由进气导流管、空气滤清器盖、空气滤清器外壳和滤芯等组成。

空气滤清器和进气管



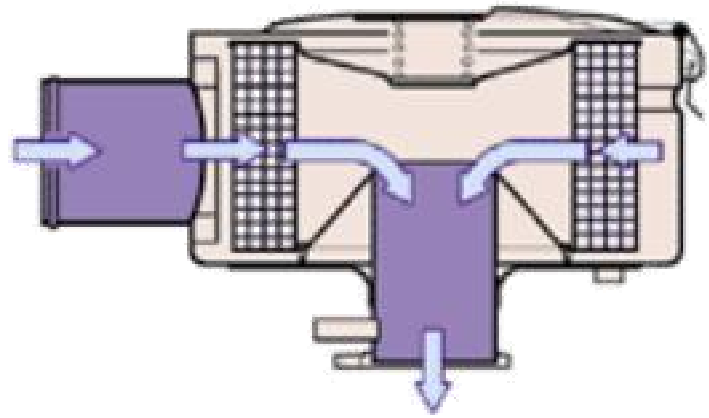
Function 功用

Ensuring sufficient clean air supply into the cylinder by separating dust

把空气中的尘土分离出来，保证供给气缸足够量的清洁空气。

Air filter overview video 视频

1. Air filter 空气滤清器



1) Inertial type 惯性式

视频(Video)

Principle 原理

Based on the principle that inertial force is proportional to masses, heavy dust can be thrown out from air during high speed rotating motion or when flow direction changes with a sudden.

它是根据离心力或惯性力与质量成正比的原理，利用尘土比空气重的特点，引导气流作高速旋转运动，重的尘土就会自动的从空气中甩出去，或着引导气流突然改变流动方向，重的尘土就会来不及改变方向而从空气中分离出去。

Advantage 优点： small intake resistance 进气阻力小
simple maintenance 保养简单

Disadvantage 缺点： poor filtration effect 滤清效果差

1. Air filter 空气滤清器

2) Filtering type 过滤式

Principle 原理

Based on the adsorption principle, filter core, such as metal net, wire, cotton or paper, separates and adsorbs dust in the airflow to filter the air.

它是根据吸附原理，引导气流通过滤蕊(如金属网、丝、棉质物质和纸质等)，将尘土隔离和粘附在滤蕊上，从而使空气得到滤清。

Advantage 优点： good filtration effect 滤清效果好

Disadvantage 缺点： large intake resistance, easily blocked
进气阻力大，易堵塞

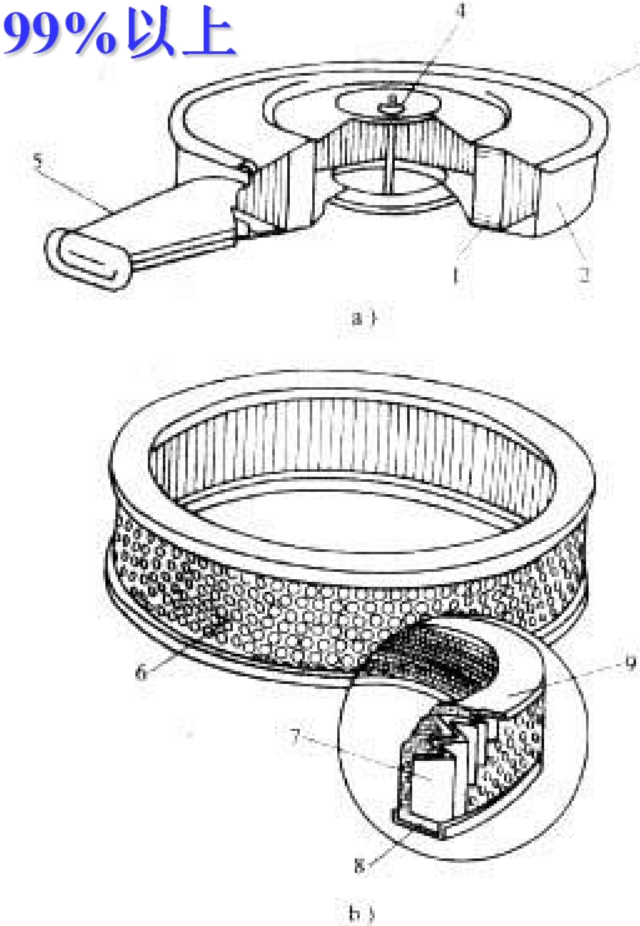
3) Oil bath filter
油浴式滤清器 效率95%-97%

油浴式空气滤清器

1-滤清器外壳 2-滤芯 3-密封圈

4-滤清器盖 5-蝶形螺母

4) Dry filter
干式滤清器 效率99%以上



干式纸滤式空气滤清器

a) 滤清器 b) 纸滤芯

1-滤芯 2-滤清器外壳 3-滤清器盖 4-蝶形螺母 5-进气导流管 6-金属网 7-打褶滤纸 8-滤芯下密封圈 9-滤芯上密封圈

1. Air fliter 空气滤清器

- Engine life will be shortened 2/3 if it is without air filter. Air filter often needs to be cleaned or renewed, or misfire during idling and poor accelerator response will occur due to blocked filter cartridge.

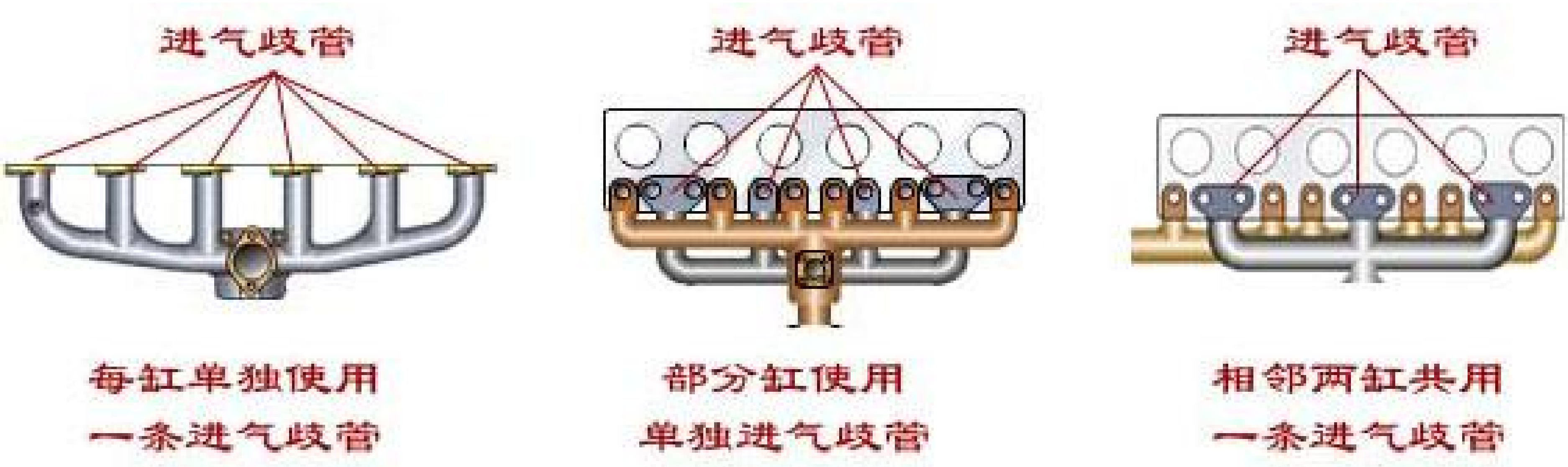
- 若不装空气滤清器，发动机寿命将缩短2/3。若空气滤清器滤芯堵塞，发动机气缸内进气不畅，怠速容易熄火，油门响应性变差（油门加大时，发动机功率变化不连续，导致车子一冲一冲的），需要经常清洗或更换。

2. Intake manifolds 进气歧管

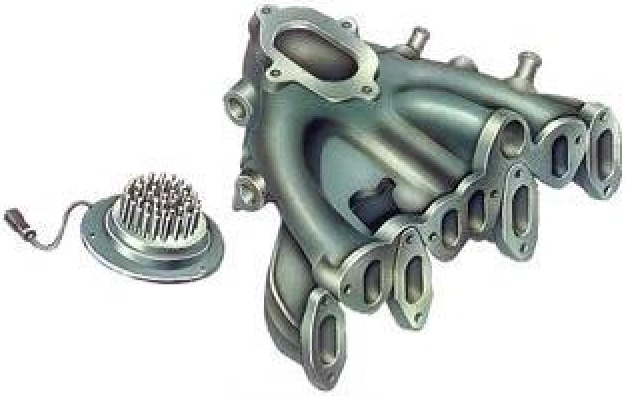
概述视频（Overview Video）

Function: Lead mixture to cylinders, ensuring cylinder-to-cylinder homogeneity.
将可燃混合气引入气缸,保证多缸机各缸进气量均匀一致.

Demand: Small intake resistance and large air inflow.
进气阻力小，充气量要大.



2. Intake manifolds 进气歧管



The length of each pipe in intake manifolds should be equal as far as possible, and its interior wall should be smooth. Alloy cast iron is applied to old type engines, and aluminum alloy is often used for car engines. In recent years, compounded plastic was also for Multiple Point Injection (MPI) engines'.

进气歧管内到各气缸的气体流道的长度尽可能相等，内壁应该光滑。一般发动机的进气歧管由合金铸铁制造，轿车发动机多用铝合金制造（重量轻，导热性好）。对于现代轿车气道喷射式（多点喷射）发动机，近年来复合塑料进气歧管也有被采用。

1) Resonant intake system 谐振进气系统

Pressure waves can be generated in intake manifolds because intake process has intermittence and periodicity. If intake valves open when positive pressure wave arrives, air inflow and engine power output will increase.

进气过程具有间歇性和周期性，因此进气支管内产生一定幅度的压力波（当地声速传播）。若利用进气支管内压力波传播的动态效应（波动效应和惯性效应），使进气门开启时正好正压力波到达进气门，则使进气充量增加，发动机功率增大.

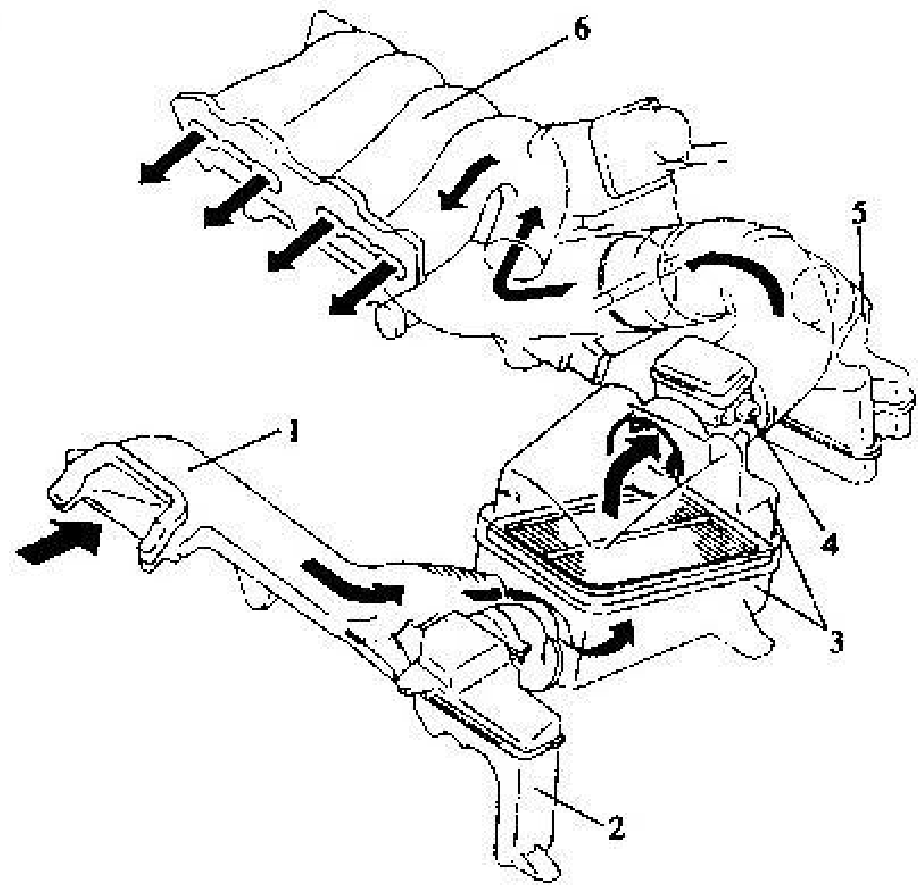


图 7-8 谐振进气系统

1-进气导流管；2-副谐振室；3-空气滤清器；4-空气流量计；5-主谐振室；6-进气歧管

2) Variable intake manifolds 可变进气歧管

In order to improve power performance and fuel economy, short and thick intake manifolds under high speed and heavy load, thin and long manifolds under low speed and load, medium manifolds under medium operating conditions should be employed.

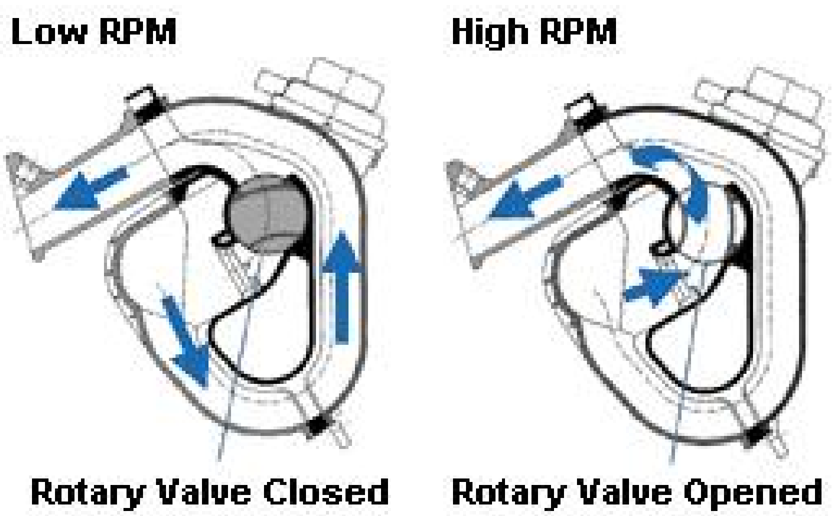
为了改善发动机的动力性和经济性，要求发动机在高转速、大负荷时装短而粗的进气支管；而在低转速、小负荷时装备细而长的进气支管；中间转速、中等负荷则居中。

2) Variable intake manifolds 可变进气歧管

1）Variable length intake manifolds 可变长度进气歧管

Low speed 低转速: Valve closes, thin and long runner to increase inflow speed and inertia. 转换阀关闭, 进气流道细而长, 提高了进气流速和气流惯性.

High speed 高转速: Valve opens, thick and short runner to decrease intake resistance. 转换阀打开，进气流道粗而细，进气阻力小.

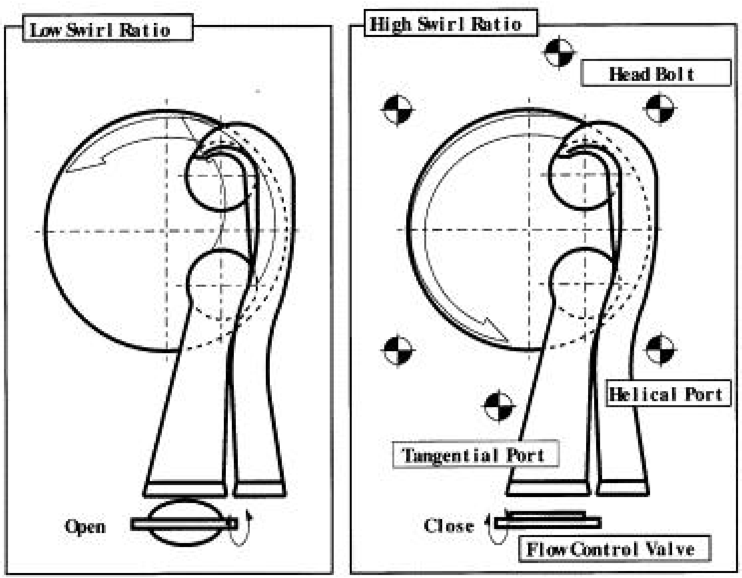


2) Variable intake manifolds 可变进气歧管

2）Variable swirl intake manifolds 可变涡流进气歧管

Low speed 低转速: Valve closes and single runner. 转换阀关闭, 空气经长进气通道进入气缸，高涡流比。

High speed 高转速: Valve opens and double runners. 转换阀打开，长进气道部分短路，空气经两个短进气道进入气缸，低涡流比。



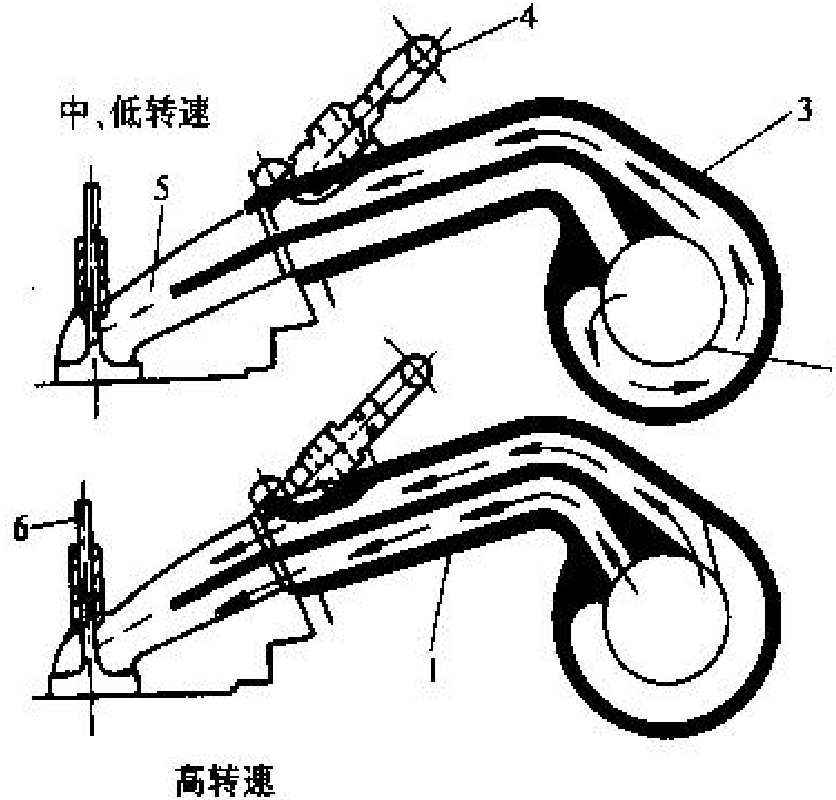


图 5-58 双通道可变进气支管

1—短进气通道 2—旋转阀 3—长进气通道
4—喷油器 5—进气道 6—进气门

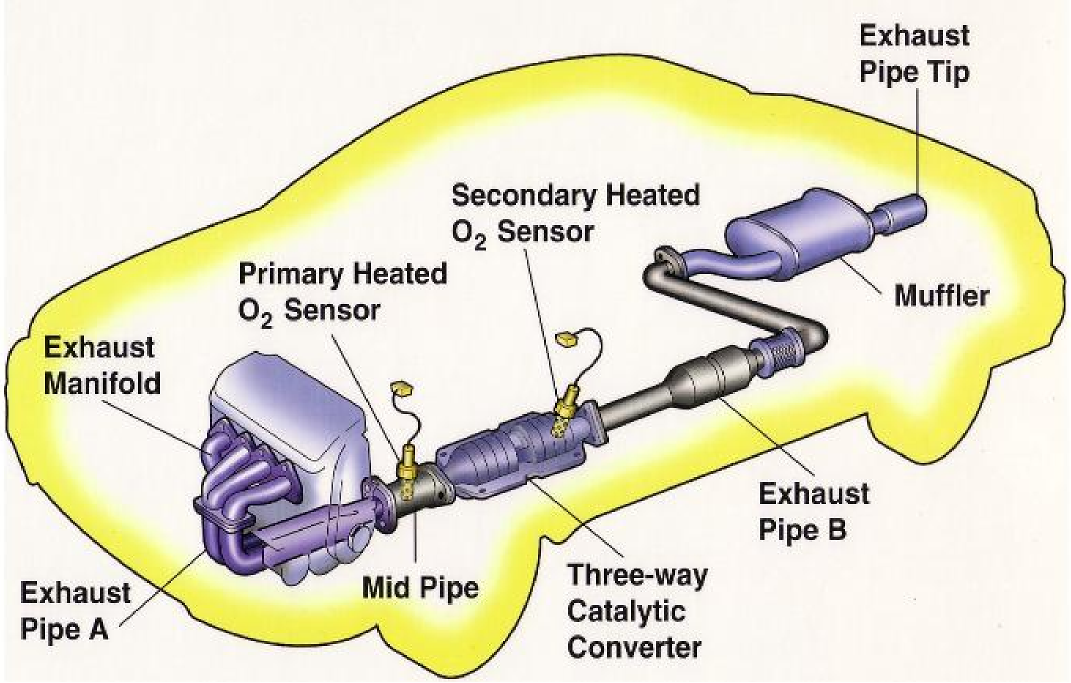
6.2 Exhaust System 排气系统

Its **function** is to exhaust residual gas after combustion. **Demand:** try to reduce exhaust resistance and noise. It mainly **consists of** exhaust manifold, exhaust pipe and muffler.

其作用是将燃烧后的废气排出；要求尽可能减少排气阻力和噪声。主要包括排气支管、排气管和消声器.

[Exhaust system overview]()

[排气系统概述（Video）]()



1. Single and Dual exhaust systems
单排气系统及双排气系统

Single exhaust system means residual gas exhausts through exhaust manifolds, exhaust pipe, catalytic converter and muffler to the atmosphere.

单排气系统指废气经排气支管、排气管、催化转换器和消声器排入大气中.

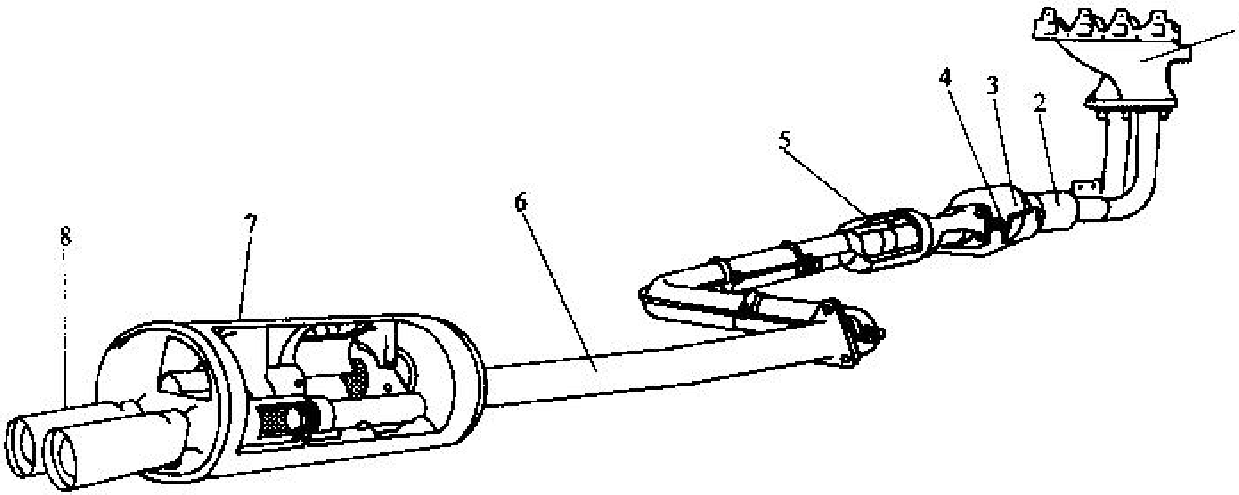


图 5-59 单排气系统的组成

1—排气支管 2—前排气管 3—催化反应器 4—排气温度传感器 5—副消声器 6—后排气管 7—主消声器 8—排气尾管

1. Single and Dual exhaust systems
单排气系统及双排气系统

V6 engines have two exhaust branch pipes, and most of them use single exhaust system, that is, using a Y-type pipe connects two exhaust branch pipes.

V6发动机有两个排气支管，大多数V6发动机采用单排气系统，即通过一个叉型管将两个排气支管连接到一个排气管上.

But some V-type engines utilize two single exhaust systems, that is, each exhaust branch pipe connects with one exhaust pipe, one catalytic converter, one muffler and one tail pipe.

但有些V型发动机采用两个单排气系统，即每个排气支管各自都连接一个排气管、催化转换器、消声器和排气尾管。

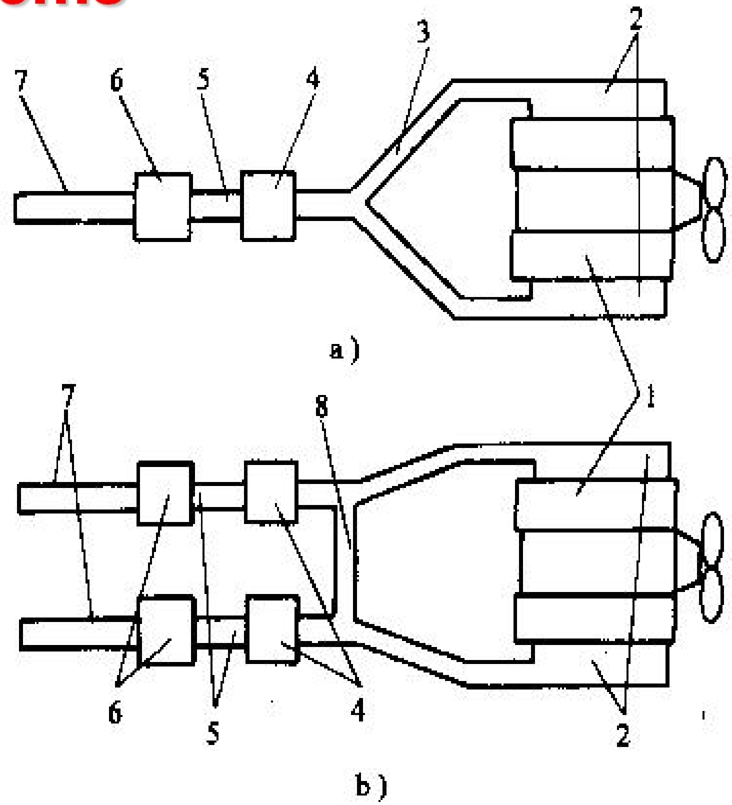


图 5-60 V 形发动机排气系统示意

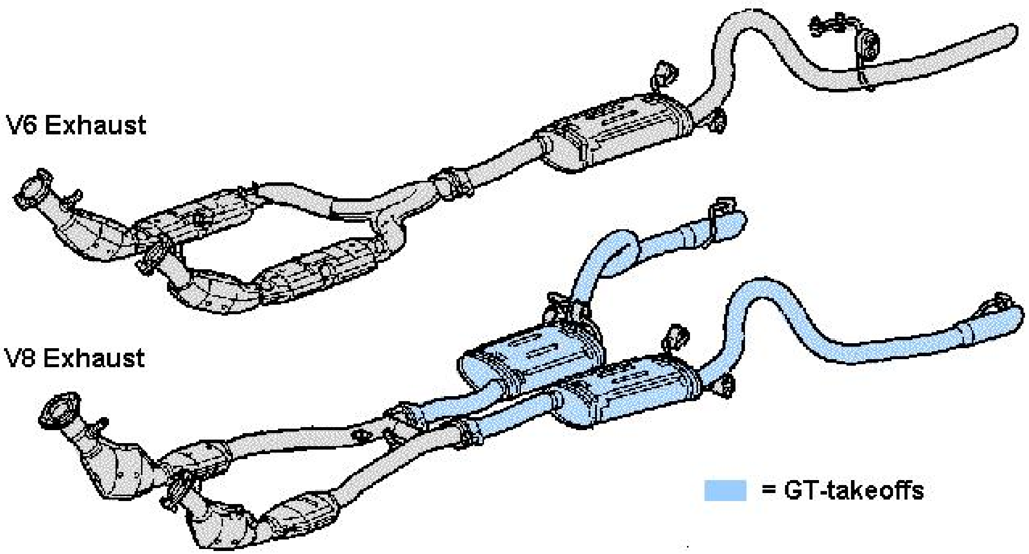
a) 单排气系统 b) 双排气系统

1—发动机 2—排气支管 3—叉形管

4—催化转换器 5—排气管 6—消声器

7—排气尾管 8—连通管

1. Single and Dual exhaust systems
单排气系统及双排气系统



2. Exhaust manifolds 排气歧管

视频（Video）

Because smooth inner wall, low resistance and light weight. Recently, more and more cars choose stainless steel as the material for exhaust manifold instead of cast iron or ductile cast iron.

一般的排气歧管由铸铁或球墨铸铁制造，近年来，采用不锈钢排气歧管的汽车愈来愈多，原因是内壁光滑，阻力小，重量轻。

2. Exhaust manifolds 排气歧管

Exhaust manifolds are made a little long to take advantage of gas inertia; they are independent of each other And have equal length. For 4-cylinder engines, branch pipes of 1^{st} and 4^{th}, 2^{nd} and 3^{rd} cylinder, which don't ignite successively, join together respectively, avoiding mutual interference.

排气歧管做得较长，为了尽可能利用气流惯性；排气支管各缸应相互独立，长度相等。四缸机分别将不连续点火的1、4及2、3缸的排气支管汇合在一起，这是为了各缸排气不出现干扰，防止出现排气倒流现象。因此，将不连续点火的气缸的排气支管汇合在一起。

2. Exhaust manifolds 排气歧管

Firing order of inline 6 cylinder engines is 1-5-3-6-2-4-1. Thus in order to get rid of interference,1,2, 3 and 4,5,6 cylinders' exhaust branch pipes should join together, respectively.

直列六缸机发火次序是1-5-3-6-2-4-1，因此，应将1、2、3三缸的排气支管以及4、5、6三缸的排气支管各自汇合在一起，可完全排除排气干扰现象

3. Exhaust Pipe 排气总管

视频（Video）

4. Muffler 消声器

Function: Draining away exhaust noise by reducing exhaust pressure and attenuating its fluctuation.

排气消声器的作用是通过降低排气压力和衰减排气压力的脉动来消减排气噪声。

4. Muffler 消声器

Muffler is made of aluminum coated steel sheet or stainless steel sheet.
消声器用镀铝钢板或不锈钢板制造。

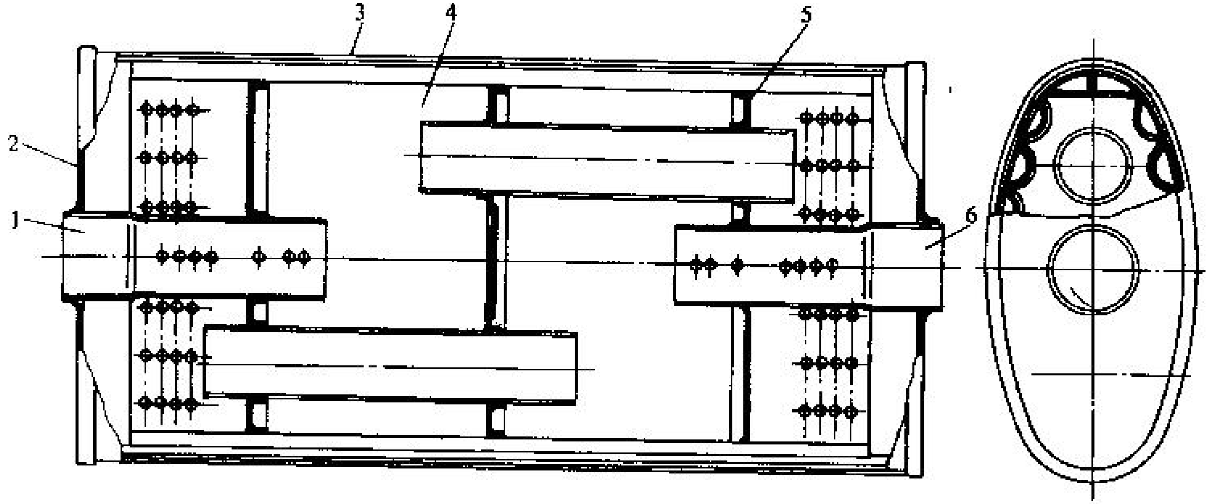


图 5-63 排气消声器结构（捷达轿车）
1—进口管 2—外隔板 3—外壳 4—内壳 5—内隔板 6—出口管

In general, muffler is composed of resonance chamber, expansion chamber and a group of porous tubes. During the process that exhaust flows through porous tubes, expansion chamber and resonance chamber, exhaust changes its flow direction constantly, gradually reducing and attenuating its pressure and fluctuation to realize energy consumption and final noise elimination.

通常，消声器由共振室、膨胀室和一组多孔的管子构成。排气经多孔的管子流入膨胀室和共振室，在此过程中，排气不断改变流动方向，逐渐降低和衰减其压力和压力脉动，消耗其能量，最终使排气噪声得到消减。

6.3 Boost System 增压系统

Definition: Provide compressed air to cylinder so as to improve air density and air inflow.将空气预先压缩后供入气缸，以提高空气密度、增加进气量的系统。

Function： Strengthen engine working process: air inflow↑ ⟶fuel delivery per cycle↑⟶ power output & thermal efficiency↑

1) Supercharger System机械增压系统

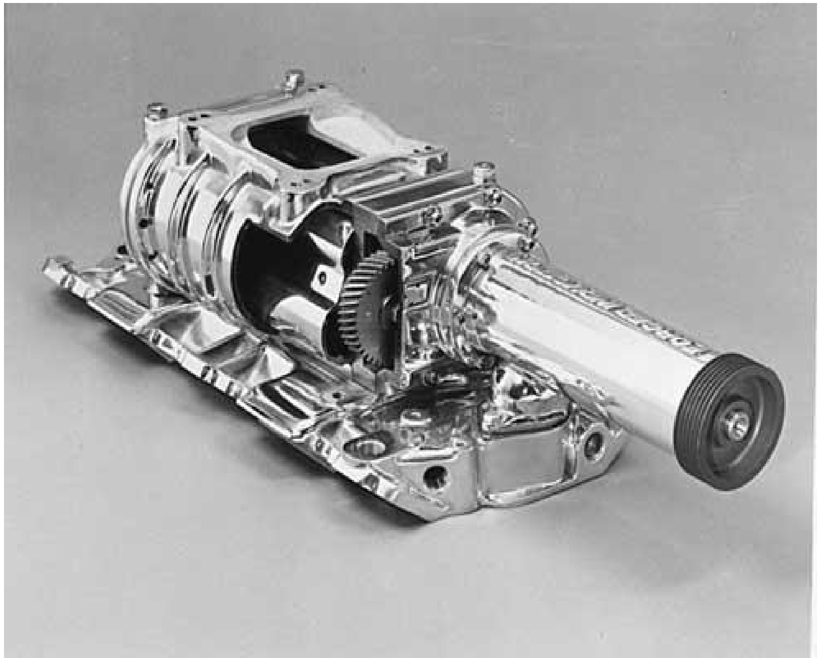
2) Turbocharger System 涡轮增压系统

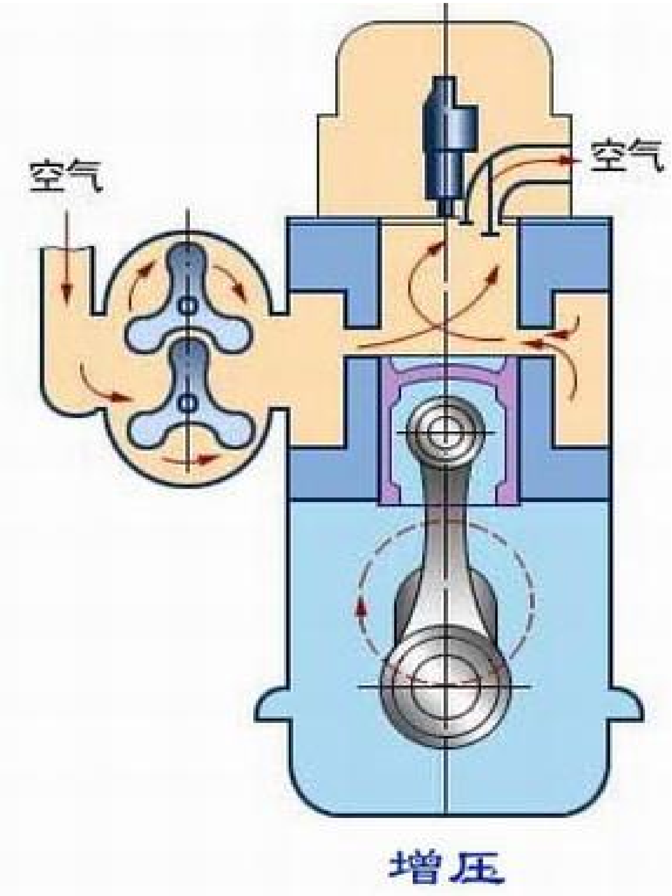
3) Pressure wave charger System气波增压系统

1) Supercharger System 机械增压系统

Supercharger applies engine rotation speed to drive its internal vane by connection between belt and crankshaft pulley, generating pressurized air into supercharging system in intake manifolds.

机械增压器采用皮带与曲轴皮带盘连接，利用发动机转动来带动机械增压器内部叶片，以产生增压空气送入进气歧管内的增压系统。



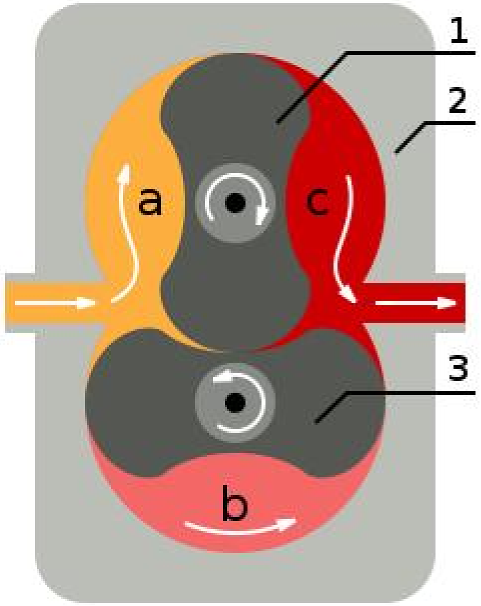


Roots-type Supercharger 罗茨式机械增压器

Principle： When rotary vane revolves, air is suctioned from compressor inlet. And the air is accelerated by the pushing of rotary vane and expressed out from the compressor outlet. The pressure ratio between outlet and inlet can be up to 1.8.

当转子旋转时，空气从压气机入口吸入，在转子叶片的推动下空气被加速，然后从压气机出口压出。出口与进口的压力比可达1.8。

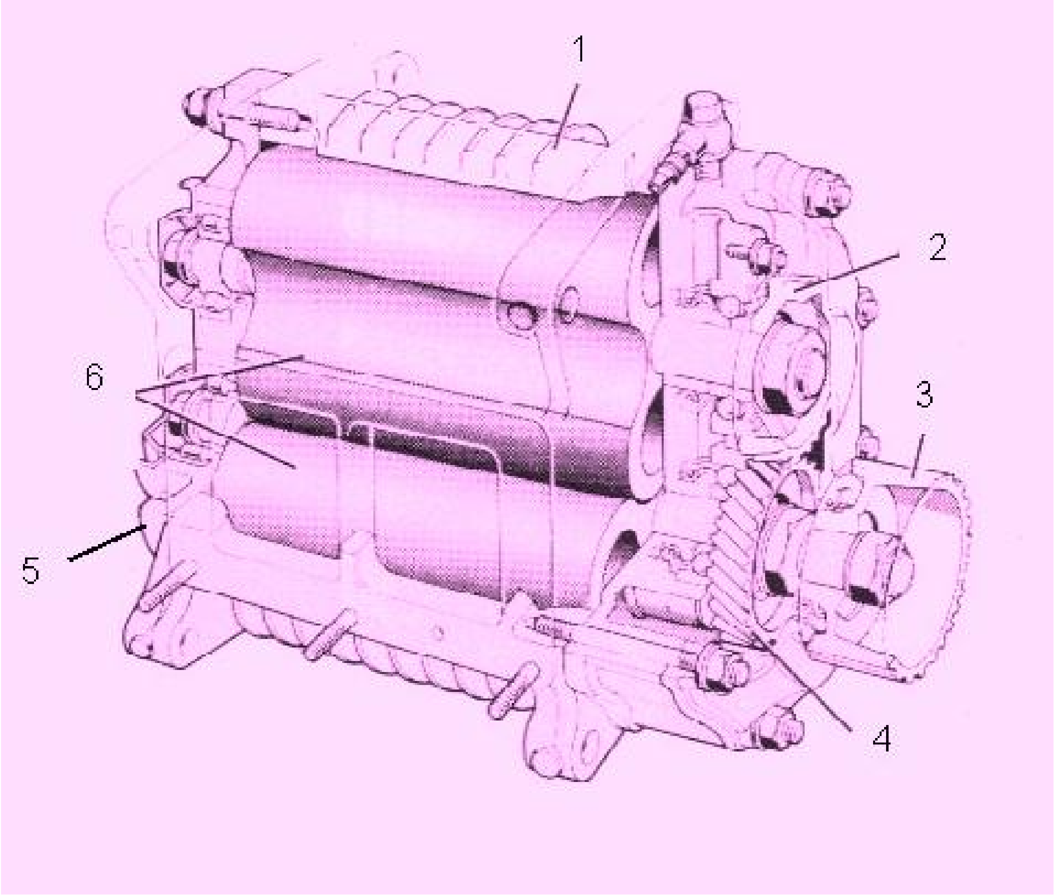
Characteristics: simple structure, high reliability, Long lasting service, air flow rate is proportional to rotation speed. 结构简单，工作可靠，使用寿命长，供气量与转速成正比。

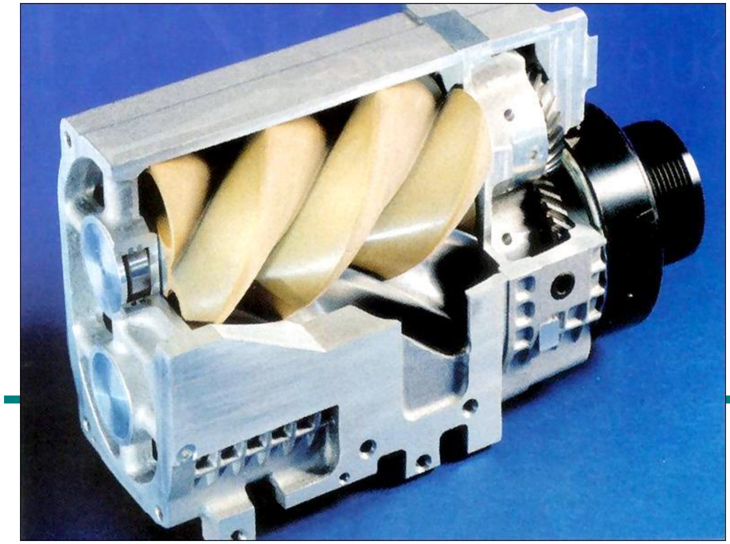


Roots-type Supercharger 罗茨式机械增压器

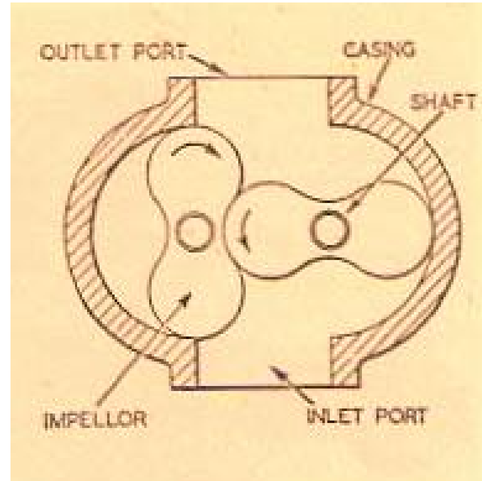
Structure:

1.shell 壳体
2.gear box 齿轮箱
3. belt pulley wheel of driving gear 传动齿轮皮带轮
4. driving gear 传动齿轮
5.electromagnetic clutch 电磁离合器
6. rotator 转子

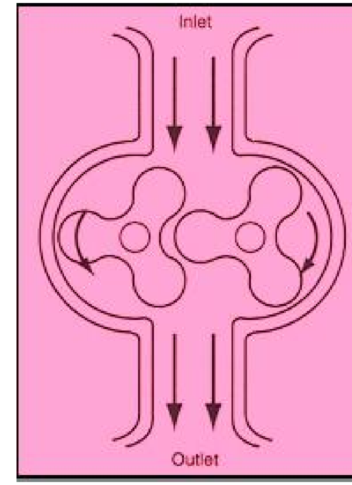




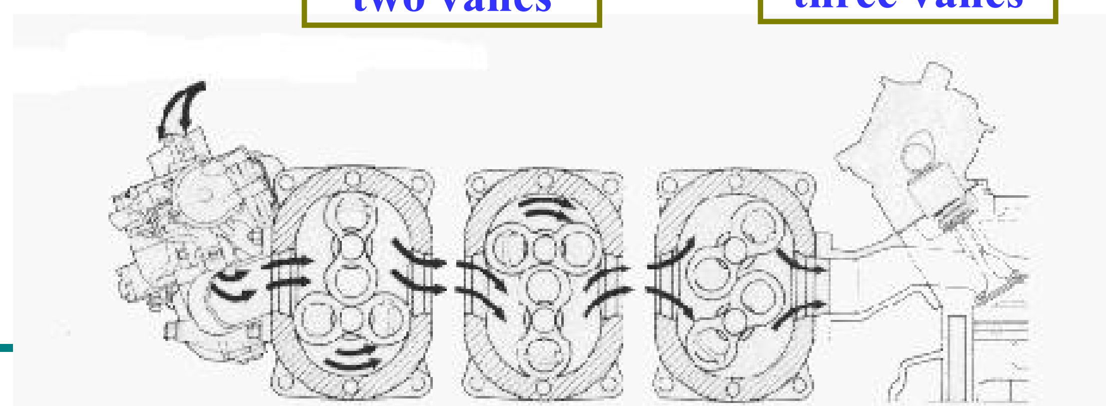
Compressor Rotator 压气机转子



两叶转子
Impellor with two vanes



三叶转子
Impellor with three vanes

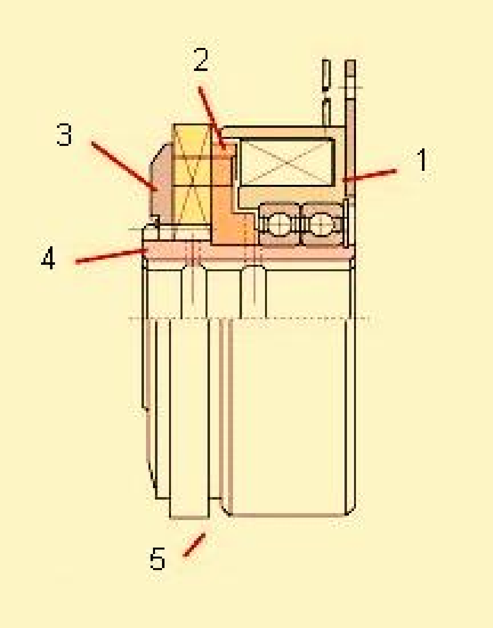


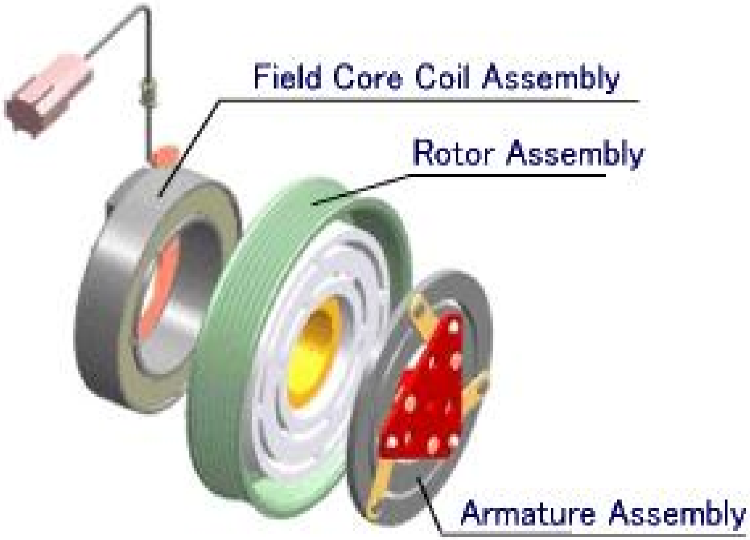
Electromagnetic Clutch 电磁离合器

Principle：According to the needs of engine operating conditions, electric unit sends out instructions to turn on or off the clutch power to control supercharger working.

电控单元根据发动机工况的需要，发出接通或切断电磁离合器电源的指令、以控制增压器的工作。当接通电源时，电磁线圈通电，主动板吸引从动摩擦片，离合器接合，增压器工作。

1.Exciting coil
励磁线圈
2.Rotator 转子
3.Armature and friction disk
衔铁和从动摩擦盘
4. inner driver
内传动装置
5.drive shaft 传动轴

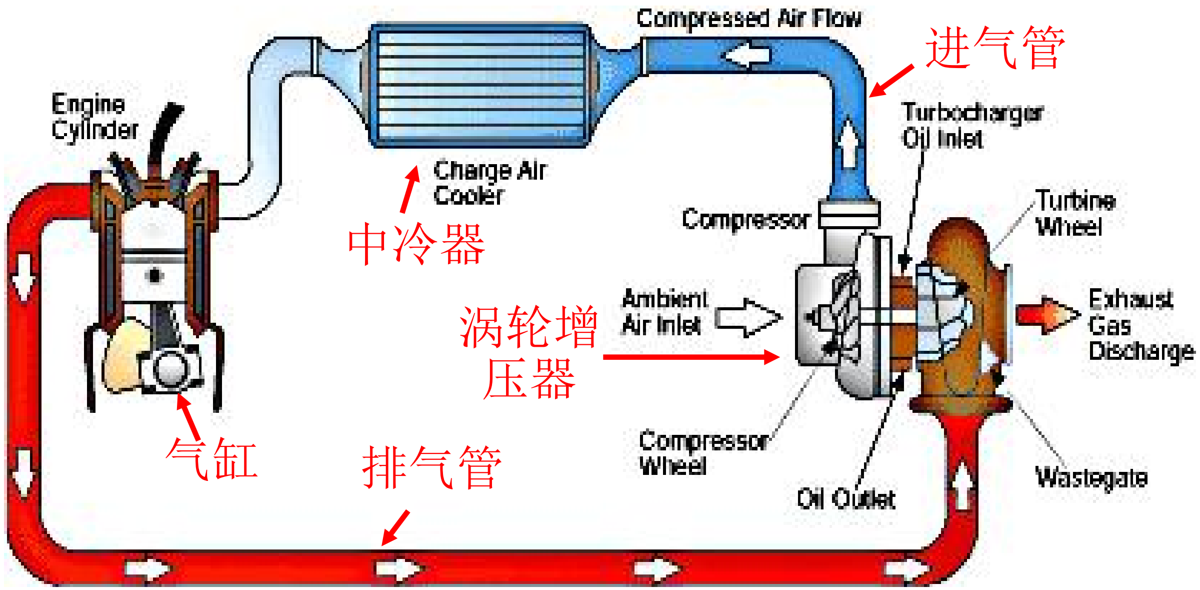




2) Turbocharger System 涡轮增压器

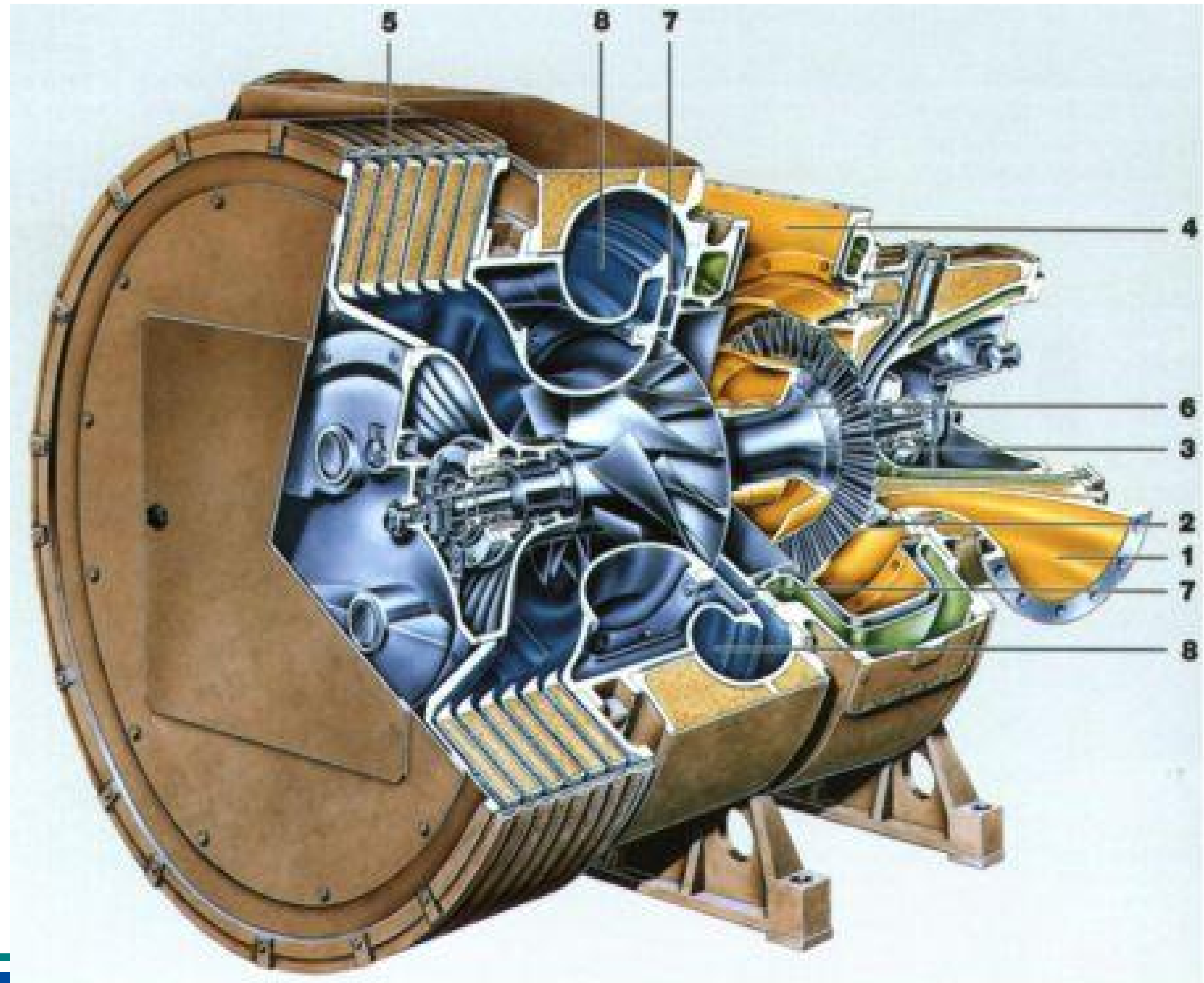
Principle： It utilizes inertial force in residual gas to push forward the turbine which will drive the coaxial vane. The vane compresses the air sent from air cleaner and let it enter cylinder.

涡轮增压器实际上是一种空气压缩机，通过压缩空气来增加进气量。它是利用发动机排出的废气惯性冲力来推动涡轮室内的涡轮，涡轮又带动同轴的叶轮，叶轮压送由空气滤清器管道送来的空气，使之增压进入气缸。



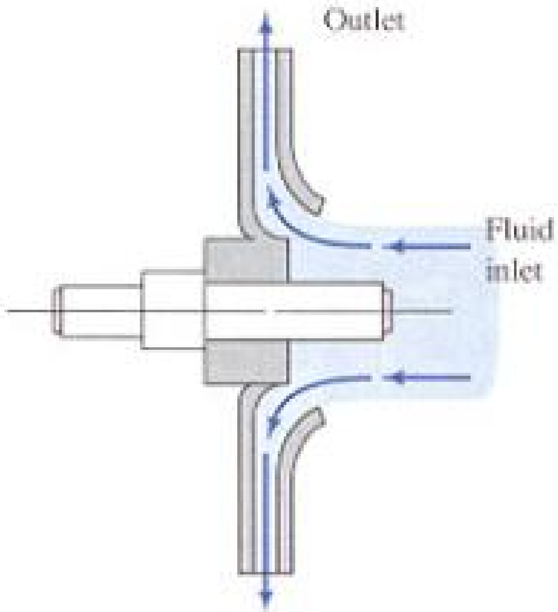
2) Turbocharger System 涡轮增压器

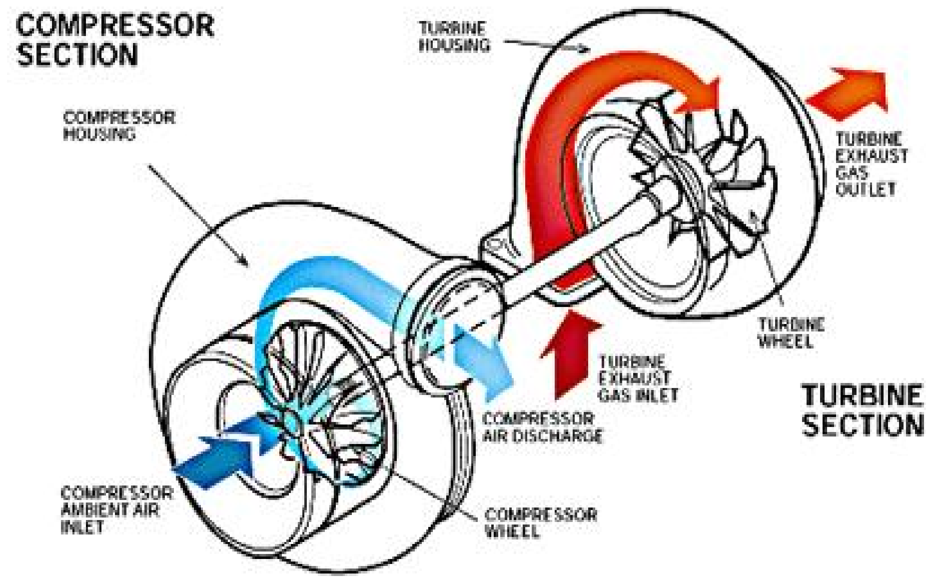
1.inlet pipe
进气管
2.turbine nozzle
涡轮机喷嘴
3.turbine impellor
涡轮机叶轮
4.Outlet pipe
排气管
5.intercooler
中冷器
6.compressor vane
压缩机叶轮
7.diffuser 扩压管
8.volute 涡壳

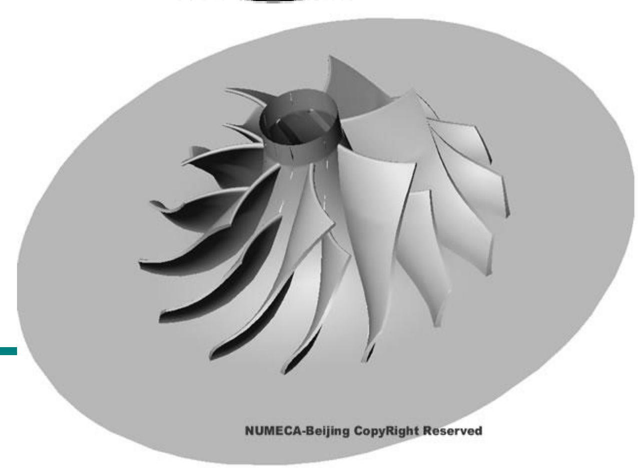


Centrifugal Compressor离心式压气机

Function： Convert mechanical energy in residual gas to intake pressure energy. 将机械能转化为进气压力能。

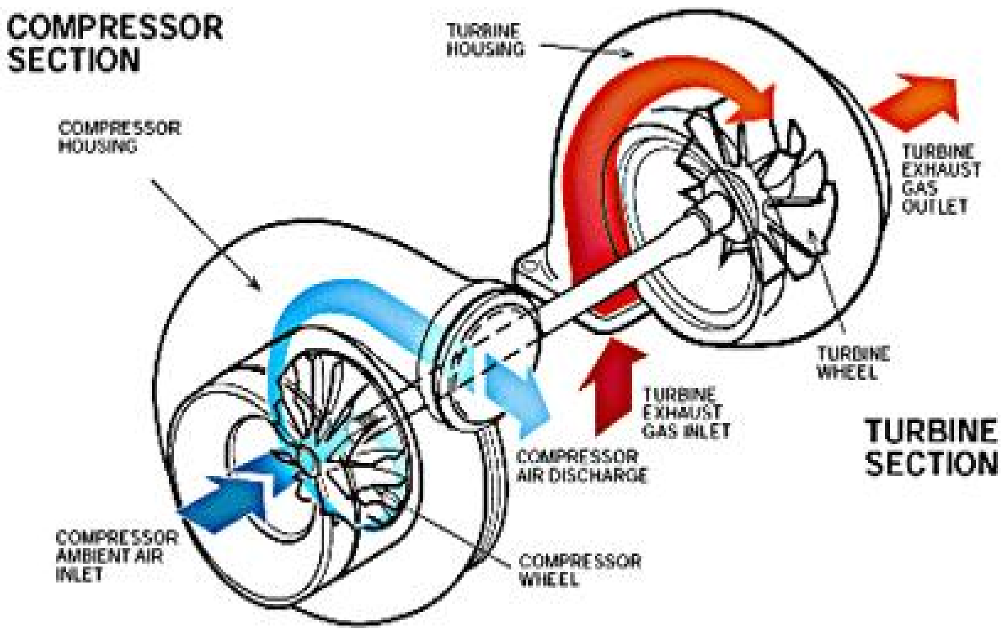


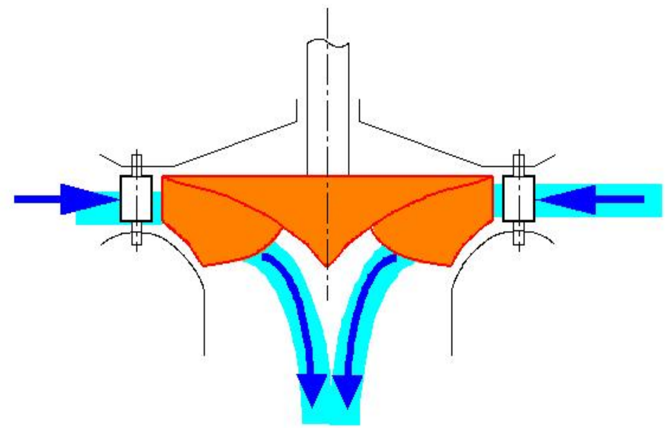




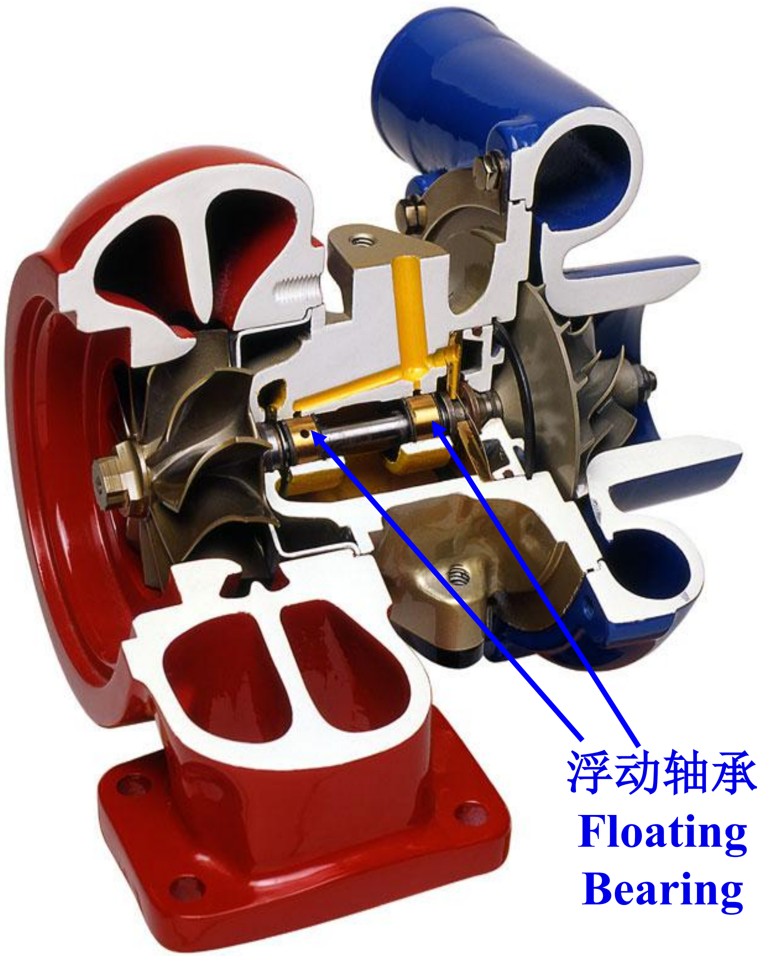
Radial Flow Turbine 径流式涡轮机

Function： Convert gas pressure in residual gas mechanical driving force.
将气体压力转化为机械传动力。





Floating Bearing 浮动轴承



Lubricating and Cooling in Turbocharger System
涡轮增压器的润滑与冷却

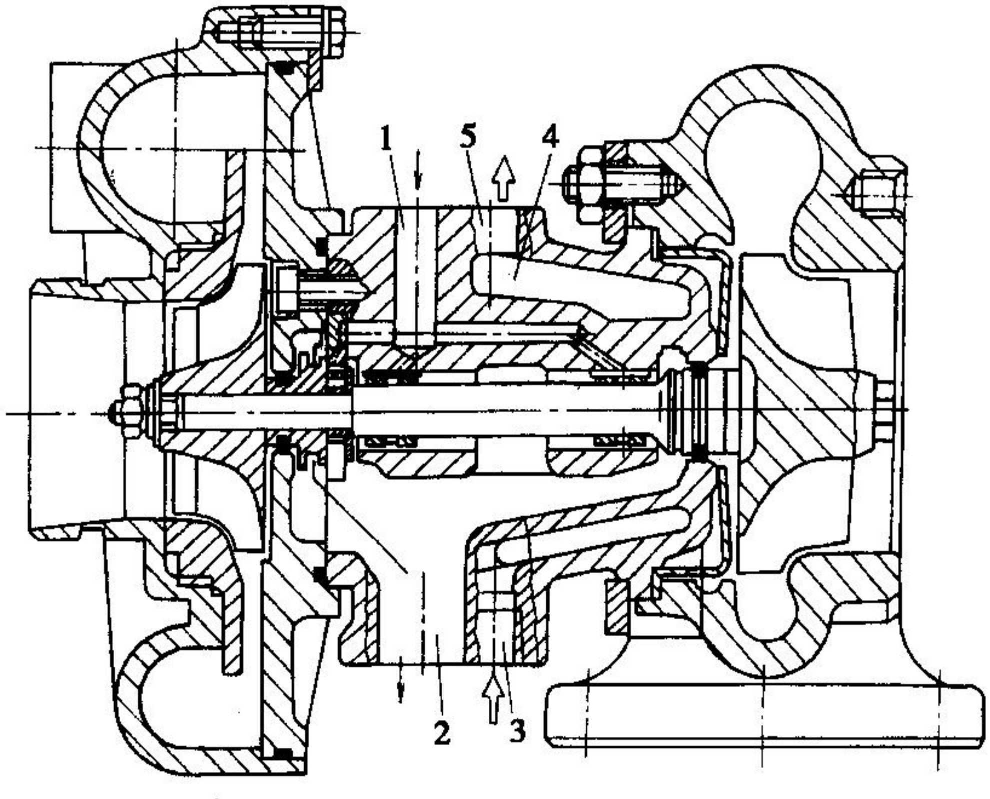
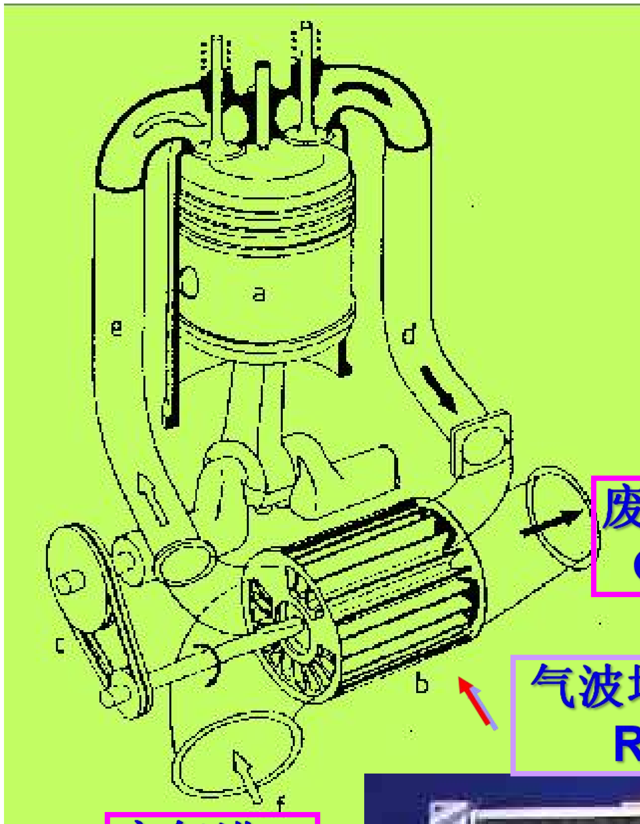


图 10-16　涡轮增压器的润滑油路及冷却水套

1-机油进口；2-机油出口；3-冷却液进口；4-冷却水套；5-冷却液出口

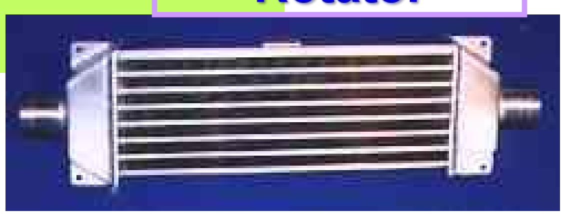
3) Pressure-wave Charger System 气波增压系统



废气出口
Outlet

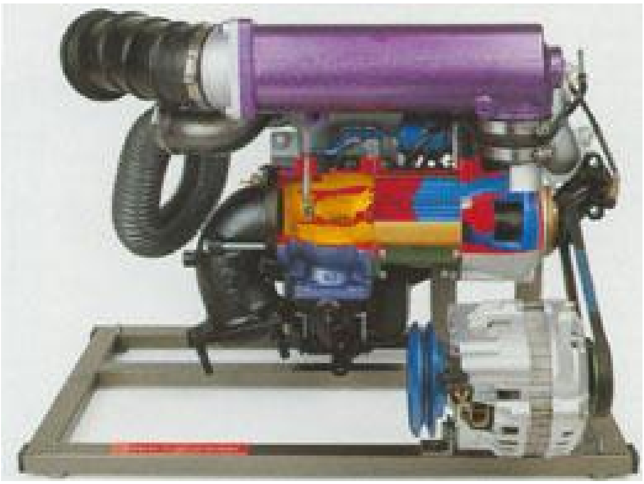
气波增压器转子
Rotator

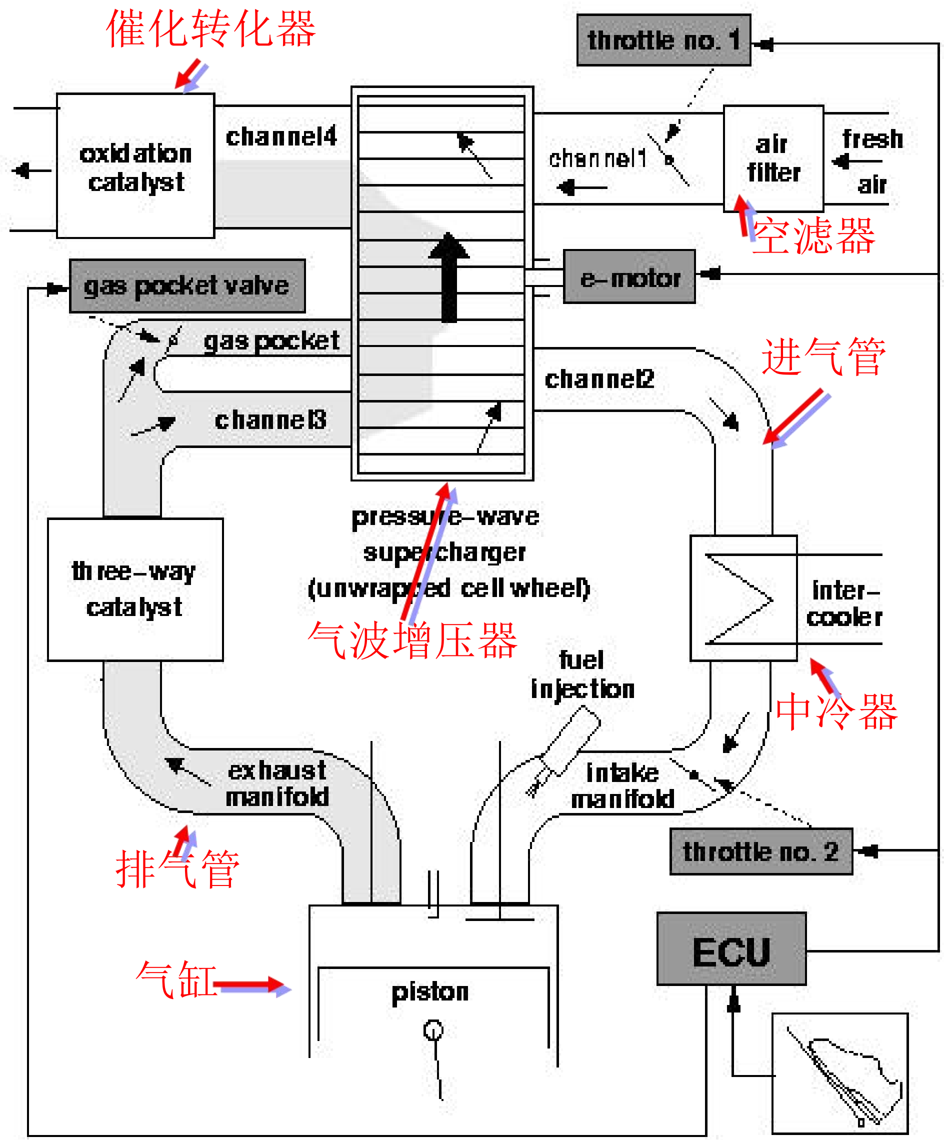
空气进口
Inlet



原理：利用进气管道中的压力波的特性，使废气和新气接触，在相互不混和的前提下，将废气能量直接传给低压空气，并提高其压力的一种方式。

3) Pressure-wave Charger System 气波增压系统





特点：结构紧凑，节省能源；但增压的范围有一定的局限性，压力波在转子流道内的传播速度只决定于排气或空气的温度，而排气温度取决于柴油机负荷，与转速无关。所以只能按柴油机某一转速来确定最佳转子尺寸和转子转速。

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